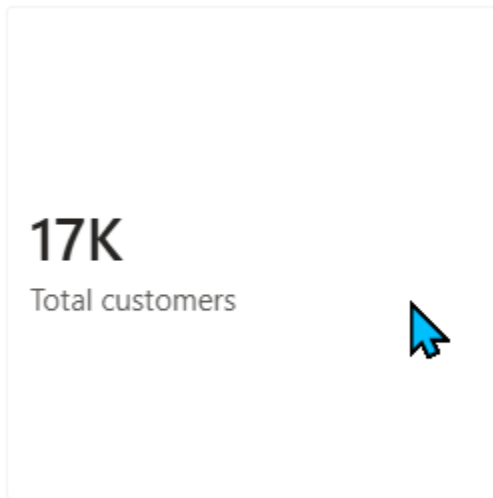
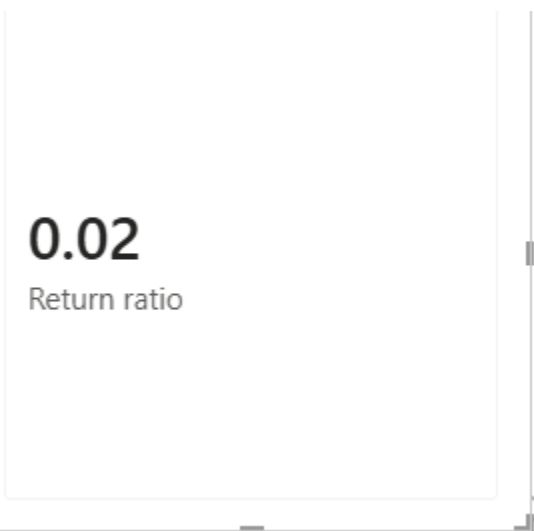


- Create a measure named Total Customers, to calculate the number of distinct AdventureWorks customers who made a transaction.
- Create a measure named Return Rate, defined as quantity returned divided by quantity sold.



Total customers = `DISTINCTCOUNT('Sales'[CustomerKey])`



Return ratio = `DIVIDE(SUM('Returns Data'[ReturnQuantity]), SUM(Sales[OrderQuantity]))`

a) Logical Function

- Create a calculated column in the Customer Lookup table named Customer Priority: ○ If the customer is a parent and has an annual income > \$100,000, Custom Priority = Priority ○ Otherwise, Customer Priority = Standard
- Create a calculated column in the Customer Lookup table named Customer income: ○ If annual income is >= \$150,000, Very High ○ If annual income is >= \$100,000, High ○ If annual income is >= \$50,000, Average ○ Otherwise, Income Level = Low
- Use a SWITCH function* to create another column named Education Category: ○ If EducationLevel is High School or Partial High School, Education Category = High School ○ If EducationLevel is Bachelors or Partial College, Education Category = Undergrad ○ If EducationLevel is Graduate Degree, Education Category = Graduate

Table: Customers (18,148 rows) Columns: customer_priority (2 distinct values)

```
customer_priority = IF(Customers[TotalChildren] >= 1 && Customers[AnnualIncome] > 100000, "Priority", "Standard")
```

Table: Customers (18,148 rows) Column: Customer_income (4 distinct values)

BirthDate	MaritalStatus	Gender	EmailAddress	AnnualIncome	TotalChildren	EducationLevel	Occupation	HomeOwner	customer_priority	Customer_income	Education Category	Birth_year	Full Name
01 April 1964	S	M	shannon38@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	SHANNON CARLSON
2 September 1964	S	M	jessie9@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	JESSIE LIU
2 November 1963	S	M	ruben1@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	RUBEN KAPOOR
9 November 1974	S	M	ruben30@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	RUBEN MUÑOZ
7 September 1965	S	M	joe14@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	JOE RANA
0 December 1963	S	M	jarrod0@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	JARROD SURI
09 October 1975	S	M	dustin20@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1975	DUSTIN GOLDSTEIN
08 July 1976	S	M	clayton29@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	CLAYTON JAI
04 March 1976	S	M	irving0@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	IRVING SCHMIDT
11 February 1974	S	M	alan10@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	ALAN HUANG
02 August 1963	S	M	brendan19@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	BRENDAN RAJI
12 April 1963	S	M	gregory24@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	GREGORY BECKER
01 March 1964	S	M	marco4@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	MARCO VANCE
08 July 1964	S	M	alejandro23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	ALEANDRO HU
0 December 1964	S	M	shane18@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	SHANE FERNANDEZ
4 November 1976	S	M	jay19@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	JAY RAMAN
01 February 1976	S	M	damen1@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	DAMEN YE
11 February 1968	S	M	roy18@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	ROY MARTINEZ
17 April 1968	S	M	pedro11@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	PEDRO RANA
02 April 1965	S	M	eugene21@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	EUGENE LIANG
14 June 1965	S	M	johnny23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	JOHNNY ANAND
9 November 1974	S	M	edwin20@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	EDWIN ZHENG
04 July 1964	S	M	carl15@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	CARL CHANDER
19 July 1976	S	M	brendan11@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	BRENDAN ANDERSEN
22 February 1965	S	M	clayton2@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	CLAYTON CHEN
7 September 1964	S	M	michal4@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	MICAH LIU
11 March 1964	S	M	troy7@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	TROY ARUN
09 February 1964	S	M	pedro31@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	PEDRO GUTIERREZ
18 January 1968	S	M	warren23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	WARREN WU
7 September 1964	S	M	ricky14@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	RICKY GILL
20 June 1975	S	M	roy8@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1975	ROY MADANI
22 August 1965	S	M	kyrie33@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	KYLE PEREZ
13 August 1974	S	M	gary16@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	GARY JIMENEZ
21 June 1963	S	M	bill15@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	BILLY GILL
21 June 1968	S	M	samuel22@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	SAMUEL HAYES
28 April 1968	S	M	connor34@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	CONNOR TURNER
7 September 1968	S	M	thomas60@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	THOMAS SMITH

```
Customer_income = SWITCH( TRUE(), Customers[AnnualIncome] >= 150000, "Very High",
Customers[AnnualIncome] >= 100000, "High", Customers[AnnualIncome] >= 50000,
"Average", "Low")
```

Table: Customers (18,148 rows) Column: Education Category (3 distinct values)

BirthDate	MaritalStatus	Gender	EmailAddress	AnnualIncome	TotalChildren	EducationLevel	Occupation	HomeOwner	customer_priority	Customer_income	Education Category	Birth_year	Full Name
01 April 1964	S	M	shannon38@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	SHANNON CARLSON
2 September 1964	S	M	jessie9@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	JESSIE LIU
2 November 1963	S	M	ruben1@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	RUBEN KAPOOR
9 November 1974	S	M	ruben30@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	RUBEN MUÑOZ
7 September 1965	S	M	joe14@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	JOE RANA
0 December 1963	S	M	jarrod0@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	JARROD SURI
09 October 1975	S	M	dustin20@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1975	DUSTIN GOLDSTEIN
08 July 1976	S	M	clayton29@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	CLAYTON JAI
04 March 1976	S	M	irving0@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	IRVING SCHMIDT
11 February 1974	S	M	alan10@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	ALAN HUANG
02 August 1963	S	M	brendan19@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	BRENDAN RAJI
12 April 1963	S	M	gregory24@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	GREGORY BECKER
01 March 1964	S	M	marco4@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	MARCO VANCE
08 July 1964	S	M	alejandro23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	ALEANDRO HU
0 December 1964	S	M	shane18@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	SHANE FERNANDEZ
4 November 1976	S	M	jay19@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	JAY RAMAN
01 February 1976	S	M	damen1@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	DAMEN YE
11 February 1968	S	M	roy18@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	ROY MARTINEZ
17 April 1968	S	M	pedro11@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	PEDRO RANA
02 April 1965	S	M	eugene21@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	EUGENE LIANG
14 June 1965	S	M	johnny23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	JOHNNY ANAND
9 November 1974	S	M	edwin20@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	EDWIN ZHENG
04 July 1964	S	M	carl15@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	CARL CHANDER
19 July 1976	S	M	brendan11@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	BRENDAN ANDERSEN
22 February 1965	S	M	clayton2@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	CLAYTON CHEN
7 September 1964	S	M	michal4@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	MICAH LIU
11 March 1964	S	M	troy7@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	TROY ARUN
09 February 1964	S	M	pedro31@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	PEDRO GUTIERREZ
18 January 1968	S	M	warren23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	WARREN WU
7 September 1964	S	M	ricky14@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	RICKY GILL

```
Education Category =
SWITCH(
    Customers[EducationLevel],
    "High School", "High School",
    "Partial High School", "High School",
    "Bachelors", "Undergrad",
    "Partial College", "Undergrad",
    "Graduate Degree", "Graduate")
```

```

        "Bachelors", "Undergrad",
        "Partial College", "Undergrad",
        "Graduate Degree", "Graduate"
    )

```

b) Text Function

- Update the Month Short column in the Calendar Lookup table to extract and capitalize the first 3 characters of the month name
- Create a new column in the Product Lookup table named SKU Category, to return any number of characters before the first hyphen in the ProductSKU column

The screenshot shows the Microsoft Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Column tools' tab is selected. The 'Name' field is set to 'Month Short'. The 'Data type' is set to 'Text'. The formula bar shows the following DAX formula:

```

1 Month Short =
2 UPPER(
3     LEFT('Calendar Lookup'[Date].[Month], 3)
4 )

```

The table below shows the data for the 'Calendar Lookup' table, with columns 'Date', 'Month Short', 'start of month', and 'Week Number'.

Date	Month Short	start of month	Week Number
01 January 2020	JAN	01-01-2020 00:00:00	1
02 January 2020	JAN	01-01-2020 00:00:00	1
03 January 2020	JAN	01-01-2020 00:00:00	1
04 January 2020	JAN	01-01-2020 00:00:00	1
05 January 2020	JAN	01-01-2020 00:00:00	2
06 January 2020	JAN	01-01-2020 00:00:00	2
07 January 2020	JAN	01-01-2020 00:00:00	2
08 January 2020	JAN	01-01-2020 00:00:00	2
09 January 2020	JAN	01-01-2020 00:00:00	2
10 January 2020	JAN	01-01-2020 00:00:00	2
11 January 2020	JAN	01-01-2020 00:00:00	2
12 January 2020	JAN	01-01-2020 00:00:00	3
13 January 2020	JAN	01-01-2020 00:00:00	3
14 January 2020	JAN	01-01-2020 00:00:00	3
15 January 2020	JAN	01-01-2020 00:00:00	3
16 January 2020	JAN	01-01-2020 00:00:00	3
17 January 2020	JAN	01-01-2020 00:00:00	3
18 January 2020	JAN	01-01-2020 00:00:00	3
19 January 2020	JAN	01-01-2020 00:00:00	4
20 January 2020	JAN	01-01-2020 00:00:00	4
21 January 2020	JAN	01-01-2020 00:00:00	4
22 January 2020	JAN	01-01-2020 00:00:00	4
23 January 2020	JAN	01-01-2020 00:00:00	4
24 January 2020	JAN	01-01-2020 00:00:00	4
25 January 2020	JAN	01-01-2020 00:00:00	4
26 January 2020	JAN	01-01-2020 00:00:00	5
27 January 2020	JAN	01-01-2020 00:00:00	5
28 January 2020	JAN	01-01-2020 00:00:00	5
29 January 2020	JAN	01-01-2020 00:00:00	5
30 January 2020	JAN	01-01-2020 00:00:00	5
31 January 2020	JAN	01-01-2020 00:00:00	5
01 February 2020	FEB	01-02-2020 00:00:00	5
02 February 2020	FEB	01-02-2020 00:00:00	6
03 February 2020	FEB	01-02-2020 00:00:00	6
04 February 2020	FEB	01-02-2020 00:00:00	6

Table: Calendar Lookup (912 rows) Column: Month Short (12 distinct values)

Month Short =

```

UPPER(
    LEFT('Calendar Lookup'[Date].[Month], 3)
)

```

File Home Help Table tools Column tools

Name SKU Category

Data type Text

Format Text

Summarization Don't summarize

Data category Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure

Formating

Properties

1 SKU Category =

2 LEFT (

3 Products[ProductSKU],

4 FIND("-", Products[ProductSKU]) - 1

5)

ProductSKU	ProductName	ModelName	ProductDescription	ProductColor	ProductSize	ProductStyle	ProductCost	ProductPrice	SKU Category
HL-U309-R	Sport-100 Helmet, Red	Sport-100	Universal fit, well-vented, lightweight, snap-on visor.	Red	0	0	13.0863	34.99	HL
HL-U309	Sport-100 Helmet, Black	Sport-100	Universal fit, well-vented, lightweight, snap-on visor.	Black	0	0	12.0278	33.6442	HL
HL-U309-B	Sport-100 Helmet, Blue	Sport-100	Universal fit, well-vented, lightweight, snap-on visor.	Blue	0	0	12.0278	33.6442	HL
LA-1098	AWC Logo Cap	Cycling Cap	Traditional style with a flip-up brim, one-size fits all.	Multi	0	U	5.7052	8.6442	CA
KL-1639	LL Fork	LL Fork	Stout design absorbs shock and offers more precise steering.	NA	0	0	65.8097	148.22	FK
KL-S136	ML Fork	ML Fork	Composite road fork with an aluminum steerer tube.	NA	0	0	77.9176	175.49	FK
KL-9939	HL Fork	HL Fork	High-performance carbon road fork with curved legs.	NA	0	0	101.8936	229.49	FK
LS-0296	LL Headset	LL Headset	Threadless headset provides quality at an economical price.	NA	0	0	15.1848	34.2	HS
LS-2451	ML Headset	ML Headset	Sealed cartridge keeps dirt out.	NA	0	0	49.1168	102.29	HS
LS-3479	HL Headset	HL Headset	High-quality 1" threadless headset with a grease port for quick lubrication.	NA	0	0	55.3801	124.72	HS
HB-M243	LL Mountain Handlebars	LL Mountain Handlebars	All-purpose bar for on or off-road.	NA	0	0	17.978	40.4909	HB
HB-M763	ML Mountain Handlebars	ML Mountain Handlebars	Tough aluminum alloy bars for downhill.	NA	0	0	24.9932	56.2909	HB
HB-M918	HL Mountain Handlebars	HL Mountain Handlebars	Flat bar strong enough for the pro circuit.	NA	0	0	48.5453	109.3364	HB
HB-R504	LL Road Handlebars	LL Road Handlebars	Unique shape provides easier reach to the levers.	NA	0	0	17.978	40.4909	HB
HB-R720	ML Road Handlebars	ML Road Handlebars	Anatomically shaped aluminum tube bar will suit all riders.	NA	0	0	24.9932	56.2909	HB
HB-R956	HL Road Handlebars	HL Road Handlebars	Designed for racers: high-end anatomically shaped bar from aluminum alloy.	NA	0	0	48.5453	109.3364	HB
WL-M423	LL Mountain Front Wheel	LL Mountain Front Wheel	Replacement mountain wheel for entry-level rider.	Black	0	0	26.9708	60.745	FW
WL-M762	ML Mountain Front Wheel	ML Mountain Front Wheel	Replacement mountain wheel for the casual to serious rider.	Black	0	0	92.8071	209.025	FW
WL-M928	HL Mountain Front Wheel	HL Mountain Front Wheel	High-performance mountain replacement wheel.	Black	0	0	123.2955	300.215	FW
WL-R623	LL Road Front Wheel	LL Road Front Wheel	Replacement road front wheel for entry-level cyclist.	Black	0	0	37.9909	85.565	FW
WL-R762	ML Road Front Wheel	ML Road Front Wheel	Sturdy alloy features a quick-release hub.	Black	0	0	110.2829	248.385	FW
WL-R820	HL Road Front Wheel	HL Road Front Wheel	Strong wheel with double-walled rim.	Black	0	0	146.5466	330.06	FW
WL-R925	Touring Front Wheel	Touring Front Wheel	Aerodynamic rim for smooth riding.	Black	0	0	96.7964	218.07	FW
WL-M423	LL Mountain Rear Wheel	LL Mountain Rear Wheel	Replacement mountain wheel for entry-level rider.	Black	0	0	38.9588	87.745	RW
WL-M762	ML Mountain Rear Wheel	ML Mountain Rear Wheel	Replacement mountain wheel for the casual to serious rider.	Black	0	0	104.7951	236.025	RW
WL-M928	HL Mountain Rear Wheel	HL Mountain Rear Wheel	High-performance mountain replacement wheel.	Black	0	0	145.2835	327.215	RW
WL-R623	LL Road Rear Wheel	LL Road Rear Wheel	Replacement road rear wheel for entry-level cyclist.	Black	0	0	49.9789	112.565	RW
WL-R762	ML Road Rear Wheel	ML Road Rear Wheel	Aluminum alloy rim with stainless steel spokes; built for speed.	Black	0	0	122.2709	275.385	RW
WL-R820	HL Road Rear Wheel	HL Road Rear Wheel	Strong rear wheel with double-walled rim.	Black	0	0	158.5346	357.06	RW
WL-R925	Touring Rear Wheel	Touring Rear Wheel	Excellent aerodynamic rim guarantee a smooth ride.	Black	0	0	108.7844	245.07	RW
PL-T100	Touring-Panniers, Large	Touring-Panniers	Durable, water-proof nylon construction with easy access. Large enough for weekend trips.	Grey	0	0	51.5625	125	PA
LO-C100	Cable Lock	Cable Lock	Wraps to fit front and rear tires, carrier and 2 keys included.	NA	0	0	10.3125	25	LO
PL-U452	Minipump	Minipump	Designed for convenience. Fits in your pocket. Aluminum barrel. 160psi rated.	NA	0	0	8.2459	19.99	PU
PL-M144	Mountain Pump	Mountain Pump	Classic and lightweight. Emergency patches stored in handle.	NA	0	0	10.1064	24.99	PU

Table: Products (293 rows) Columns: SKU Category (40 distinct values)

SKU Category =

```
LEFT (
    Products[ProductSKU],
    FIND("-", Products[ProductSKU]) - 1
)
```

c) Date function

- Create a new column in the Customer Lookup table named Birth Year, to extract only the year from the BirthDate column.

File Home Help Table tools Column tools

Name Birth_year
Data type Whole number
Format \$ - % 0
Summarization Sum
Data category Uncategorized
Sort by column
Data groups
Manage relationships
New column

Structure
1 Birth_year = YEAR(Customers[BirthDate])

BirthDate	MaritalStatus	Gender	EmailAddress	AnnualIncome	TotalChildren	EducationLevel	Occupation	HomeOwner	customer_priority	Customer_Income	Education Category	Birth_year	Full Name
01 April 1964	S	M	shannon38@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	SHANNON CARLSON
September 1964	S	M	jessie9@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	JESSIE LIU
November 1963	S	M	ruben1@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	RUBEN KAPOOR
November 1974	S	M	ruben30@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	RUBEN MAUROZ
September 1965	S	M	joe14@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	JOE RANA
December 1963	S	M	jarrod0@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	JARROD SURI
09 October 1975	S	M	dustin20@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1975	DUSTIN GOLDSTEIN
08 July 1976	S	M	clayton29@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	CLAYTON JAI
04 March 1976	S	M	irving0@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	IRVING SCHMIDT
11 February 1974	S	M	alan10@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	ALAN HUANG
02 August 1963	S	M	brendan10@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	BRENDAN RALJ
12 April 1963	S	M	gregory24@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	GREGORY BECKER
01 March 1964	S	M	marco4@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	MARCO VANCE
08 July 1964	S	M	alejandro23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	ALEANDRO HU
0 December 1964	S	M	shane18@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	SHANE FERNANDEZ
November 1976	S	M	jay19@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	JAY RAMAN
21 February 1976	S	M	damen7@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	DAMEN YE
17 February 1968	S	M	roy18@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	ROY MARTINEZ
12 April 1968	S	M	pedro11@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	PEDRO RANA
02 April 1965	S	M	eugene21@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	EUGENE LIANG
14 June 1965	S	M	johnny23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	JOHNNY ANAND
November 1974	S	M	edwin20@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	EDWIN ZHENG
04 July 1964	S	M	carl15@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	CARL CHANDER
17 July 1976	S	M	brendan11@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1976	BRENDAN ANDERSEN
22 August 1965	S	M	clayton2@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	CLAYTON CHEN
September 1964	S	M	micah14@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	MICAH LIU
11 March 1964	S	M	troj7@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	TROY ARUN
29 February 1964	S	M	pedro31@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	PEDRO GUTIERREZ
18 January 1968	S	M	warren23@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	WARREN WU
September 1964	S	M	ricky14@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1964	RICKY GILL
20 June 1975	S	M	roy8@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1975	ROY MADANI
22 August 1965	S	M	kye33@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1965	KYLE PEREZ
13 August 1974	S	M	gary16@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1974	GARY JIMENEZ
21 June 1963	S	M	billy15@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1963	BILLY GILL
21 June 1968	S	M	samuel22@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	SAMUEL HAYES
28 April 1968	S	M	connor34@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	CONNOR TURNER
September 1968	S	M	thomas60@adventure-works.com	70000	0	Bachelors	Professional	N	Standard	Average	Undergrad	1968	THOMAS SMITH

Table: Customers (18,148 rows) Column: Birth_year (70 distinct values)

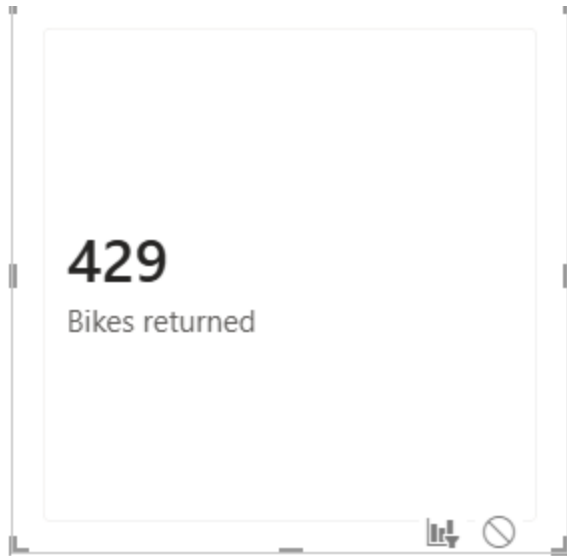
Data

- Search
- Total Revenue
- Calendar Lookup
- Date
- Date Hierarchy som
- Month Short
- start of month
- Week Number
- Customers
- AnnualIncome
- Birth_year
- BirthDate
- Customer_income
- customer_priority
- CustomerKey
- Education Category
- EducationLevel
- EmailAddress
- FirstName
- Full Name
- Gender
- HomeOwner
- LastName
- MaritalStatus
- Occupation
- Prefix
- TotalChildren
- Product Categories Lookup
- Product Subcategories Lookup
- Products
- ModeName
- ProductColor
- ProductCost

Birth_year = YEAR(Customers[BirthDate])

d) Calculate Function

- Create a new measure named Bike Returns to calculate the total quantity of bikes returned
- Create a matrix to show Bike Returns (values) by Start of Month (rows). What do you notice about the volume of bike returns over time?
- Create a new measure named Bike Sales to calculate the total quantity of bikes sold, and add it to the matrix. What do you notice?
- Create a new measure named Bike Return Rate using either CALCULATE or DIVIDE, and add it to the matrix

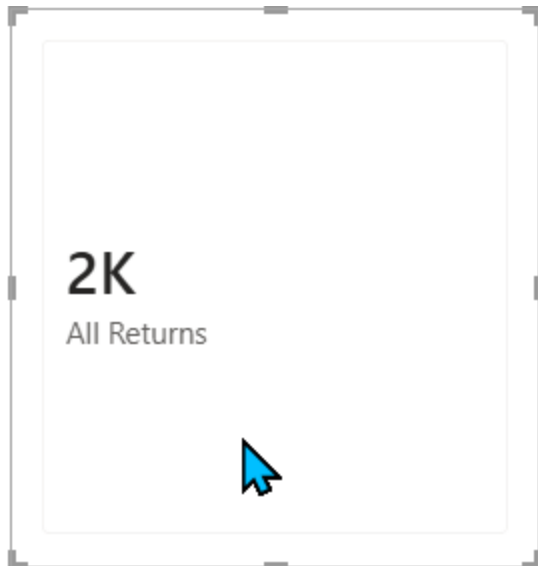


Bikes returned = `CALCULATE(SUM('Returns Data'[ReturnQuantity]), 'Product Categories Lookup'[ProductCategoryKey] = 1 )`

Month	Bikes returned	Bikes_sold	Bike Return Rate
January	26	1192	0.02
February	34	1238	0.03
March	44	1352	0.03
April	57	1450	0.04
May	57	1651	0.03
June	46	1681	0.03
July	15	753	0.02
August	20	763	0.03
September	24	771	0.03
October	37	835	0.04
November	30	879	0.03
December	39	1364	0.03
Total	429	13929	0.03

e) Calculate & all function

- Create a new measure named All Returns to calculate the total number of returns, regardless of filter context
- Create a new measure named % of All Returns that divides Total Returns by All Returns
- Create a matrix to show % of All Returns (values) by product Category Name (rows).
- Which category accounts for the largest percentage of returns? The smallest?"



All Returns = `CALCULATE( SUM('Returns Data'[ReturnQuantity]) , ALL('Product Categories Lookup'))`

Total Returns	All Returns	% of All Returns
1828	1828	100.00

% of All Returns =

```
DIVIDE(
    [Total Returns],
    [All Returns]
) * 100
```

CategoryName	Total Returns	All Returns	% of All Returns
Accessories	1130	1828	61.82
Bikes	429	1828	23.47
Clothing	269	1828	14.72
Components		1828	
Total	1828	1828	100.00

f) Iterator Function

- Create a new measure named Total Cost that multiplies the order quantities in the Sales Data table by the product cost in the Product Lookup table, then calculates the sum
- Create a new measure named Total Profit (revenue minus cost)
- Create a matrix to show Total Profit (values) by Year (rows).
- How much profit has AdventureWorks earned so far in 2022

Total Cost =

```
SUMX(  
    Sales,  
    Sales[OrderQuantity] * RELATED(Products[ProductCost])  
)
```

Total Profit =

```
[Total Revenue] - [Total Cost]
```

Year	Total Profit
2020	26,01,602.33
2021	39,67,084.13
2022	38,89,028.97
Total	1,04,57,715.43



g) Time Intelligence Function

Add the following measures to the model:

- Previous Month Returns
- Previous Month Orders
- Previous Month Profit
- Order Target (10% increase over previous month)
- Profit Target (10% increase over previous month)
- 90-day Rolling Profit

start of month	Total Returns	Previous Month Returns
01-01-2020 00:00:00	4	
01-02-2020 00:00:00	4	4
01-03-2020 00:00:00	9	4
01-04-2020 00:00:00	14	9
01-05-2020 00:00:00	11	14
01-06-2020 00:00:00	4	11
01-07-2020 00:00:00	3	4
01-08-2020 00:00:00	6	3
01-09-2020 00:00:00	2	6
01-10-2020 00:00:00	11	2
01-11-2020 00:00:00	5	11
01-12-2020 00:00:00	13	5
Total	1828	

Previous Month Returns =

```

CALCULATE(
    [Total Returns],
    PREVIOUSMONTH('Calendar Lookup'[Date])
)

```

Order Target =

```

[Previous Month Orders] + ([Previous Month Orders] * 0.10)

```

start of month	Total Orders	Previous Month Orders	Order Target
01-01-2020 00:00:00	184		
01-02-2020 00:00:00	165	184	202.40
01-03-2020 00:00:00	198	165	181.50
01-04-2020 00:00:00	204	198	217.80
01-05-2020 00:00:00	206	204	224.40
01-06-2020 00:00:00	212	206	226.60
01-07-2020 00:00:00	247	212	233.20
01-08-2020 00:00:00	278	247	271.70
01-09-2020 00:00:00	196	278	305.80
01-10-2020 00:00:00	223	196	215.60
01-11-2020 00:00:00	191	223	245.30
01-12-2020 00:00:00	326	191	210.10
Total	84174		

Previous Month Orders =

```
CALCULATE(  
    [Total Orders],  
    PREVIOUSMONTH('Calendar Lookup'[Date])  
)
```

start of month	Total Profit	Previous Month Profit	Profit Target
01-01-2020 00:00:00	2,35,814.03		
01-02-2020 00:00:00	2,12,186.69	2,35,814.03	2,59,395.43
01-03-2020 00:00:00	2,59,084.52	2,12,186.69	2,33,405.36
01-04-2020 00:00:00	2,63,031.34	2,59,084.52	2,84,992.97
01-05-2020 00:00:00	2,66,275.75	2,63,031.34	2,89,334.48
01-06-2020 00:00:00	2,70,067.51	2,66,275.75	2,92,903.33
01-07-2020 00:00:00	1,96,682.79	2,70,067.51	2,97,074.26
01-08-2020 00:00:00	2,18,355.47	1,96,682.79	2,16,351.07
01-09-2020 00:00:00	1,40,516.15	2,18,355.47	2,40,191.02
01-10-2020 00:00:00	1,68,581.76	1,40,516.15	1,54,567.76
01-11-2020 00:00:00	1,34,175.98	1,68,581.76	1,85,439.94
01-12-2020 00:00:00	2,36,830.33	1,34,175.98	1,47,593.58
Total	1,04,57,715.43		

Previous Month Profit =

```
CALCULATE(  
    [Total Profit],  
    PREVIOUSMONTH('Calendar Lookup'[Date])  
)
```

Profit Target =

```
[Previous Month Profit] * 1.10
```

start of month	Total Profit	90-Day Rolling Profit
01-01-2020 00:00:00	2,35,814.03	2,35,814.03
01-02-2020 00:00:00	2,12,186.69	4,48,000.72
01-03-2020 00:00:00	2,59,084.52	7,03,629.47
01-04-2020 00:00:00	2,63,031.34	7,34,302.55
01-05-2020 00:00:00	2,66,275.75	7,72,192.20
01-06-2020 00:00:00	2,70,067.51	7,94,581.77
01-07-2020 00:00:00	1,96,682.79	7,11,349.52
01-08-2020 00:00:00	2,18,355.47	6,63,634.33
01-09-2020 00:00:00	1,40,516.15	5,47,909.27
01-10-2020 00:00:00	1,68,581.76	5,15,402.27
01-11-2020 00:00:00	1,34,175.98	4,42,034.38
01-12-2020 00:00:00	2,36,830.33	5,28,612.75
Total	1,04,57,715.43	21,42,663.11

```
90-Day Rolling Profit =  
CALCULATE(  
    [Total Profit],  
    DATESINPERIOD(  
        'Calendar Lookup'[Date],  
        MAX('Calendar Lookup'[Date]),  
        -90,  
        DAY  
    )  
)
```